Certified Answers as an Audit

Five Provably-False Keys on Humanity's Last Exam, and the Discipline That Makes Them Worth Reporting

Matthew Parslow
Independent Researcher

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Abstract

A system that answers a question only when it can certify the answer, and otherwise returns an honest miss, is not a competitor on a benchmark. It is an *auditor* of one. Run across the text-only portion of Humanity's Last Exam (Phan et al., 2025), deriving every answer before the key is read, it certified 43 answers: 36 confirm their keys and 5 convict them, with no accusation withdrawn and one of its own earlier flags retracted after re-derivation found a sign error in the system rather than in the key. The five convictions are set out with their diagnoses. *Coverage is* 1.8% *and that is the honest headline*: this is not a result about how much the system can answer, it is a result about what certified answering makes visible when it is pointed at the key rather than at the score.

1 Why a frontier benchmark must contain errors

This is arithmetic and not an accusation, and stating it first prevents the rest reading as one.

A benchmark of some 2,370 research-grade questions is authored at the pointy end of every field it covers, which is exactly where its reviewers stand at the edge of their own knowledge. At any plausible post-review error rate the expected number of wrong keys exceeds one. *The instrument built to test reasoning is bound by the same arithmetic of error rates as everything else*, and a benchmark with no wrong keys would be the surprising object, not this.

What follows is therefore not a complaint about the benchmark, which is a serious piece of work whose difficulty is the reason it is useful here. It is an argument that a certain kind of answering system is the right instrument for finding the errors any such benchmark must have.

2 Certification turns answering into auditing

The system answers only where it can derive the answer under an independent certificate — a second derivation by a structurally different route, a held-exact computation, or a conservation or limiting case forced by the stem. Where it cannot, it returns a miss. *The key is read only afterwards, and only to compare.*

That ordering is what makes the arrangement an audit. Every certified answer puts one key on trial and returns one of two verdicts: it confirms the key, or it convicts it. A conviction is banked only when the derivation is certified sound, the key's error is demonstrable, and no reading of the stem rescues the key — the last checked adversarially, by trying to rescue it.

Remark 2.1 (Zero false accusations is the whole of the value). An audit claiming three convictions with airtight proofs of each is worth more than one claiming thirty. The discipline is therefore asymmetric on purpose: a disagreement is not a conviction, and a sound-but-rescuable disagreement is recorded and *not* banked (Section 4).

The system also retracts against itself. One earlier physics flag was withdrawn when rederivation located a sign error in the system rather than in the key. *An auditor that has never withdrawn anything has not been audited*, and the withdrawal is reported here for that reason rather than omitted as untidy.

3 The five convictions

Each is stated as: what the key says, what was certified, and where the key's reasoning goes wrong. The diagnosis matters more than the disagreement — a conviction that cannot say *why* the key is wrong is only a second opinion.

Mathematics — rank of a Hurewicz kernel (key 28, certified 95). For $Y = X_1 \# X_2 \# X_3$ with $\pi_1(Y) = \mathbb{Z}/5 * \mathbb{Z}/8 * \mathbb{Z}/2$, the kernel of abelianisation is free by Kurosh, and its rank is 95 by three independent routes: an orbifold Euler characteristic ($\chi(K) = 80 \cdot (-47/40) = -94$, rank $= 1 - \chi(K)$), a Schreier deficiency count, and an explicit degree-80 cover ($V = 80$, $E = 240$, $F = 66$). *The key computed* $\prod(n_i - 1) = 28$*, which is the correct kernel rank only for* ***two*** *free factors* — verified against $[5, 8] \to 28$, and against a divergent control $[2, 2, 2]$ where the construction gives the correct 5 and the key's formula gives 1.

Physics — sudden expansion of an infinite well (key 4.03%, certified 0%). The ground state of a well of width L with the right wall moved *instantaneously* to $3L$. Under a sudden change the state vector does not have time to evolve, so the expectation of the energy is unchanged. *A decrease of* 4.03% *requires* $\langle E \rangle$ *to change, which the word "instantaneous" in the stem forbids.*

Chemistry — contracted basis functions for toluene in 6-31G (key 54, certified 79). Counting per atom under 6-31G gives 9 for carbon and 2 for hydrogen, so $7 \cdot 9 + 8 \cdot 2 = 79$. *The key's* 54 *corresponds to no coherent count* — it matches none of the standard countings of the same basis.

Mathematics — domino tilings of an Aztec diamond (key 6, certified 8). The tiling count of an Aztec diamond of order n is $2^{n(n+1)/2}$ (Elkies et al., 1992), so it is a power of two for every n: 2, 8, 64, 1024, and so on. *Six is not a power of two.* The certified answer of 8 is the order-two count, obtained by two independent routes. This is the cleanest of the five, because refuting the key requires only the parity of a theorem and no recomputation of the question at all.

Computer science — true boolean expressions of length five (key 44, certified 47). Two structurally independent routes — filtering all 7^5 strings through a recursive-descent parser, and generating bottom-up from the grammar — both yield 90 valid expressions of which 47 are true, on the identical string set. *The key is self-refuting*: the setter's own enumeration includes two expressions that evaluate to false while filed as true, and omits five genuinely true expressions of equal standing. It retains both redundant parentheses and repeated negation, so no consistent grammar excludes what it excludes, and no precedence convention yields 44 — standard precedence gives 47, the opposite convention gives 43.

4 What was not convicted, and why that section matters more

Six further questions produced a sound internal value disagreeing with the key, and none was banked, because in each case a reading of the stem rescues the key. *These are the cases that make the five above worth anything.*

They include an underdetermined projectile problem where the explosion energy is unspecified, a liquid-mirror exponent that is regime-dependent on an unstated rest depth, an α-range question reproducible under one range law but not the textbook one, and two convention- or rounding-dependent values.

The sharpest is a graph-theoretic fixed-point count where an exhaustive search over the standard atlas finds *seven* against the key's six, and the sole difference is the single-vertex graph K_1 — which is connected, and is a fixed point, and which the setter's own code excludes by filtering degree-zero vertices. *That exclusion is defensible, so no conviction was recorded*, even though the literal reading of the stem gives seven. An audit willing to accuse here would be worth less everywhere.

5 What this is not

Coverage is 43 *questions of* 2,370*, which is* 1.8%. The system declines the overwhelming majority of the benchmark, and nothing here should be read as a claim about how much of it can be answered. A benchmark score and a certified audit are different measurements and the second is not a weak version of the first.

Nor is any of it a claim about the reasoning of systems in general. The faculties are general rather than per-question — no hand-coding, no answer-baking, no reverse-engineering from keys — but the sample that reaches certification is selected precisely by being certifiable, which is a strong and non-random condition. *The right reading is that certified coverage doubles as audit at no additional cost*, and the audit is what is reported.

6 Falsifiability commitments

1. [SHAPE] **Zero false accusations.** Fails on the exhibition of a rescuing reading for any one of the five, in which case that conviction is withdrawn and the count is reported lower. Each is stated with its diagnosis specifically so that this is attemptable.

2. [DETAIL] **The certificate is independent of the key.** Fails if any certified value can be shown to depend on the key having been read, which the ordering is designed to prevent and which the record is kept to make checkable.

3. [SHAPE] **The diagnoses locate the error.** Fails where a key is wrong for a different reason than the one given — the conviction survives and the explanation does not, which is a distinct and weaker failure and is reported as such.

4. [DETAIL] **Tier-B is not a holding pen.** Fails if any Tier-B item is later convicted on evidence available at the time it was set aside, which would mean the bar was applied inconsistently rather than conservatively.

5. [SHAPE] **Coverage is not the claim.** This paper commits to the negative: no statement about benchmark performance is made or implied, and a reader who extracts one has found an error rather than a result.

References

Noam Elkies, Greg Kuperberg, Michael Larsen, and James Propp. Alternating-sign matrices and domino tilings (Part I). *Journal of Algebraic Combinatorics*, 1(2):111–132, 1992.

Long Phan et al. Humanity's Last Exam. 2025.